

Table 1. Primitives for Defect Specification

Symbol	Semantics
Static Analysis (Slither)	
$\text{methods}(c)$	The set of public/external methods defined in contract c .
$\text{IsReceive}(m)$	Holds if m corresponds to the Solidity <code>receive()</code> function.
$\text{IsFallback}(m)$	Holds if m corresponds to the Solidity <code>fallback()</code> function.
$\text{IsCritical}(\sigma)$	Holds if the operation executed in state σ is critical (e.g., <code>selfdestruct</code>).
$\text{YieldsControl}(c, c', \sigma)$	Holds if, in state σ , contract c executes an instruction (typically <code>CALL</code> , <code>STATICCALL</code> , or <code>DELEGATECALL</code>) that transfers execution flow to an external contract c' , creating a window for a callback.
$\text{Influence}(U, V)$	Holds if the value V is mathematically dependent on inputs provided by an external user U . It is used to identify “control gaps” in arithmetic operations.
$\text{Logic}(\phi)$	Represents the set of all semantic guards and validation logic implemented within the contract that are intended to enforce the truth of the predicate ϕ along a specific execution path.
Runtime Verification (LLM-generated based on specific template)	
$\text{Interm}(\sigma_i, \sigma, m)$	Holds if σ_i is an intermediate state for the execution of method m starting from state σ
$\text{Exec}(m, \sigma, \sigma')$	Holds if invoking m in state σ successfully transitions to σ' .
$\text{FailExec}(m, \sigma, \sigma')$	Holds if invoking m in state σ its execution fails in state σ' .
$\text{Spec}(m, \sigma)$	Holds if the functional specification of m is satisfied in state σ .
$\text{Profit}(\sigma)$	Holds if the attacker’s balance or internal accounting shares in state σ have increased relative to their initial deposit, indicating a successful exploit of state inconsistency.
$\text{Authorized}(\text{Role}, m)$	A mapping that identifies which roles are conceptually intended to access method m . A vulnerability exists if $\mathcal{P}$ grants access to an address that does not map to an authorized <i>Role</i> .
$\text{IsPanic}(\sigma)$	Holds if a state transition resulted in a “Panic” revert (Solidity 0.8.x), which corresponds to internal logic errors (e.g., assert failure or array out-of-bounds) rather than user-side input validation (require).
$\text{dGas}(\pi, \text{EIP})$	A path-integral function representing the cumulative gas cost of an execution path π under specific protocol rules (e.g., EIP-2929). It explicitly models the increased cost of “Cold” storage access.
$\text{IsGriefable}(m)$	Holds if the method, when forced to fail for all users, result in a permanent Denial-of-Service or locked funds.
$\text{IsEssential}(m)$	Holds if a method is vital for the contract’s primary functional purpose, such that their failure renders the contract “unusable” or “economically unviable”.
$\text{IsCriticalDecision}(v)$	Holds if a value v is used as the primary seed for a state transition with high financial impact (e.g., determining a lottery winner or NFT rarity).